

Disproof of Conjecture 00000001090: there is no MDS code of type $[12, 6]$ over $\text{GF}(3)$

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2026-10-04

Abstract

Conjecture 00000001090 asserts that the maximal order of the automorphism group of an MDS code over $\text{GF}(3)$ of type $[12, 6]$ is 660, attained by an M_{11} embedding, and that the orders for all other equivalence classes divide 132. We prove that *no* ternary MDS code of type $[12, 6]$ exists. More generally, if a $k \times n$ matrix over $\mathbb{F}_3$ with $k \geq 2$ has all nonzero combinations of its rows of weight at least $n - k + 1$, then $n \leq k + 2$. Hence no code attains the order 660, and the conjecture is false whatever notion of automorphism group is used. If the word “MDS” is dropped, the claim is still false: the ternary $[12, 6]$ code of repeated pairs has $2^6 \cdot 6! = 46080 > 660$ permutation automorphisms. The main results are formalized in Lean 4 (core library only).

1 The conjecture and how we read it

Definition: The automorphism group of an MDS code over $\text{GF}(3)$ of type $[12, 6]$ is the stabilizer of its permutation equivalence class. *Conjecture:* The maximal order of this automorphism group is 660 (attained by an M_{11} embedding); the group orders of all other equivalence classes divide 132.

The Chinese version says the same.

A ternary linear code of type $[n, k]$ is a k -dimensional subspace $C \subseteq \mathbb{F}_3^n$. Its minimum distance d is the least Hamming weight of a nonzero codeword. The Singleton bound gives $d \leq n - k + 1$, and C is *MDS* (maximum distance separable) if $d = n - k + 1$. So an MDS code of type $[12, 6]$ is a ternary $[12, 6, 7]$ code.

The conjecture is a conjunction. Its first clause says that the maximum of $|\text{Aut}(C)|$ over MDS $[12, 6]$ ternary codes C is 660, and that this value is *attained*. In particular it asserts that there is an MDS $[12, 6]$ ternary code whose automorphism group has order 660. We refute this clause, which refutes the conjecture. The remaining clauses are about all MDS $[12, 6]$ ternary codes. Since there are none, they are vacuously true, and we make no claim about them.

The definition of the automorphism group is worded unusually. The usual meaning is the stabilizer of C in the group acting on codes: permutations of coordinates (the permutation group $\text{PAut}(C)$), or monomial maps (permutations combined with nonzero scalars, giving $\text{MAut}(C)$). Over $\text{GF}(3)$ there are no nontrivial field automorphisms, so MAut is also the full automorphism group. Our main result is that the set of MDS $[12, 6]$ ternary codes is empty, so the refutation does not depend on which notion is meant (Section 4).

2 Notation

Write $\mathbb{F}_3 = \{0, 1, 2\}$ with arithmetic mod 3. Every nonzero $x \in \mathbb{F}_3$ satisfies $x^2 = 1$. For $w \in \mathbb{F}_3^n$, $\text{wt}(w)$ is the number of nonzero coordinates. A $k \times n$ matrix G with rows $g_0, \dots, g_{k-1}$ sends a message $m \in \mathbb{F}_3^k$ to the codeword $mG = \sum_i m_i g_i$. The rows are linearly independent if $mG \neq 0$ for all $m \neq 0$. In that case $C = \{mG\}$ is an $[n, k]$ code. Every $[n, k]$ code arises this way (take a basis as the rows).

3 The main theorem

Lemma 1 (linear dependence). *Let $r < k$ and let $v_0, \dots, v_{k-1} \in \mathbb{F}_3^r$. There is a nonzero $c \in \mathbb{F}_3^k$ with $\sum_i c_i v_i = 0$.*

Proof. Induction on r . For $r = 0$ take $c = (1, 0, \dots, 0)$. Let $r \geq 1$ and look at the last vector $w = v_{k-1}$. If $w = 0$, take $c = (0, \dots, 0, 1)$. Otherwise choose a coordinate j_0 with $p = w_{j_0} \neq 0$, and put $v'_i = v_i - (v_{i,j_0} p) w$ for $i < k-1$. Since $p^2 = 1$, every v'_i vanishes at j_0 . Deleting coordinate j_0 gives $k-1 > r-1$ vectors of $\mathbb{F}_3^{r-1}$. By induction there is a nonzero $c' \in \mathbb{F}_3^{k-1}$ with $\sum_i c'_i v'_i = 0$. Then $c = (c'_0, \dots, c'_{k-2}, -\sum_i c'_i v_{i,j_0} p)$ is nonzero and $\sum_i c_i v_i = \sum_i c'_i v'_i = 0$. $\square$

Lemma 2 (counting). *For $x, y \in \mathbb{F}_3$ let $N(x, y)$ be the number of the eight pairs $(a, b) \in \mathbb{F}_3^2 \setminus \{(0, 0)\}$ with $ax + by \neq 0$. Then $N(0, 0) = 0$ and $N(x, y) = 6$ otherwise.*

Proof. If $(x, y) \neq (0, 0)$, the map $(a, b) \mapsto ax + by$ is a nonzero linear form on $\mathbb{F}_3^2$. Its kernel has 3 elements, one of which is $(0, 0)$. So $9 - 3 = 6$ of the eight nonzero pairs give a nonzero value. $\square$

Theorem 3. *Let $k \geq 2$ and let G be a $k \times n$ matrix over $\mathbb{F}_3$ such that every nonzero message m gives $\text{wt}(mG) \geq n - k + 1$. Then $n \leq k + 2$.*

Proof. Suppose $n \geq k + 3$.

Step 1. Apply Lemma 1 to the restrictions of the rows $g_0, \dots, g_{k-1}$ to the first $k-2$ coordinates. This gives a nonzero message $m^{(1)}$ such that $u = m^{(1)}G$ vanishes on coordinates $0, \dots, k-3$. Fix i_0 with $m_{i_0}^{(1)} \neq 0$.

Step 2. Apply Lemma 1 with $r = k-1$ to the vectors $(g_{i,0}, \dots, g_{i,k-3}, [i = i_0]) \in \mathbb{F}_3^{k-1}$. Here the last entry is 1 for $i = i_0$ and 0 otherwise. This gives a nonzero message $m^{(2)}$ such that $v = m^{(2)}G$ vanishes on coordinates $0, \dots, k-3$, and $m_{i_0}^{(2)} = 0$.

Step 3. Let $(a, b) \neq (0, 0)$. The message $m = am^{(1)} + bm^{(2)}$ is nonzero: if $a \neq 0$ then $m_{i_0} = am_{i_0}^{(1)} \neq 0$; if $a = 0$ then $b \neq 0$ and $m = bm^{(2)} \neq 0$. Hence $\text{wt}(au + bv) = \text{wt}(mG) \geq n - k + 1$. Summing over the eight nonzero pairs and exchanging the sums,

$$8(n - k + 1) \leq \sum_{(a,b) \neq (0,0)} \text{wt}(au + bv) = \sum_{j=0}^{n-1} N(u_j, v_j) \leq 6(n - k + 2),$$

by Lemma 2, because $u_j = v_j = 0$ for the first $k-2$ coordinates. So $2(n - k) \leq 4$, that is $n \leq k + 2$, a contradiction. $\square$

Theorem 3 is the case $q = 3$ of the classical bound $n \leq q + k - 1$ for MDS codes with $k \geq 2$ [1]. In coding-theoretic language, Steps 1–2 shorten the code $k-2$ times, producing a two-dimensional code of length $n - k + 2$ with minimum distance at least $n - k + 1$. The counting of Step 3 is the Plotkin argument for that two-dimensional code. The bound is sharp: the tetracode generated by $(1, 0, 1, 1)$ and $(0, 1, 1, 2)$ is a $[4, 2, 3]$ MDS code.

Corollary 4. *There is no ternary $[12, 6, 7]$ code. More strongly, no 6×12 matrix over $\mathbb{F}_3$ has all nonzero combinations of its rows of weight ≥ 7 . Likewise, there is no ternary $[12, 7, 6]$ code.*

Proof. Take $(n, k) = (12, 6)$, respectively $(12, 7)$, in Theorem 3: $12 > k + 2$. $\square$

Concretely, for $[12, 6]$: there are 9 codewords vanishing on the first four coordinates, forming a 2-dimensional space. Restricted to the last 8 coordinates, the 8 nonzero ones would all need weight ≥ 7 , total ≥ 56 . But each of the 8 coordinates contributes at most 6, so the total is at most 48. The Griesmer bound gives another proof: a ternary $[n, 6, 7]$ code needs $n \geq \sum_{i=0}^5 \lceil 7/3^i \rceil = 14$.

4 Consequences for the conjecture

Theorem 5. *Conjecture 00000001090 is false.*

Proof. By Corollary 4 there is no MDS code over $\text{GF}(3)$ of type $[12, 6]$. So no such code has an automorphism group of order 660, and the first clause (“the maximal order is 660, attained”) fails. $\square$

This argument covers every reasonable reading:

- *Notion of automorphism group.* Permutation group, monomial group, full automorphism group, or literally “the stabilizer of the permutation equivalence class”: none of them matter, because there is no code. In Lean we state the conjecture for PAut and for MAut . We also prove that no MDS $[12, 6]$ ternary code has *any* property P .
- *Definition of MDS.* We only use $d \geq n - k + 1 = 7$, so the strict ($d = 7$) and non-strict readings are both covered.
- *Type $[12, 6]$ as $[n, d]$.* If $[12, 6]$ meant length 12 and minimum distance 6, an MDS code would be $[12, 7, 6]$, which does not exist either (Corollary 4).
- *“MDS” dropped.* Suppose the claim is read for all ternary $[12, 6]$ codes, for example because the author had the ternary Golay code in mind. It is still false. The code of repeated pairs

$$P = \{(x_0, x_0, x_1, x_1, \dots, x_5, x_5)\}$$

is a $[12, 6, 2]$ code. Its weight-2 words have exactly the supports $\{2i, 2i + 1\}$, so every automorphism permutes these six pairs. Conversely, every permutation preserving the pair partition is an automorphism. So $|\text{PAut}(P)| = 2^6 \cdot 6! = 46080 > 660$. Lean exhibits 768 distinct permutation automorphisms of P . For the monomial group the order is even larger, since $\text{PAut} \subseteq \text{MAut}$. (The ternary Golay code $[12, 6, 6]$ is also well known to have monomial automorphism group $2.M_{12}$ of order 190080 [2]. We cite this but do not verify it.)

Remark 6. The statement is also internally inconsistent: $|M_{11}| = 7920$, so M_{11} does not embed in a group of order 660. ($660 = |\text{PSL}(2, 11)|$, and $\text{PSL}(2, 11)$ is a maximal subgroup of M_{11} .) We do not use this.

Remark 7 (non-vacuity). The obstruction is exactly the distance 7. Ternary $[12, 6]$ codes exist, for example the extended ternary Golay code with generator matrix $[I_6 \mid A]$, where the rows of A are 011111, 101221, 110122, 121012, 122101, 112210. Its weight distribution is 1, 264, 440, 24 in weights 0, 6, 9, 12, so its minimum distance is $6 = n - k$, one less than MDS. MDS codes themselves exist in the admissible range $n \leq k + 2$ (the tetracode). Lean proves both facts.

5 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project (core library only, no Mathlib). $\mathbb{F}_3$ is `Fin 3`.

- `comb k c G j =  $\sum_{i < k} c_i G_{ij}$` , `codeword k G m`, `wt n w` (Hamming weight on coordinates $< n$), `Nonzero k m`. `IsGenMatrix n k G`: the rows are linearly independent (an $[n, k]$ code). `InCode`, `MinDistGe`, `HasMinDist`, and `IsMDS n k G := IsGenMatrix  $\wedge$  minimum distance exactly  $n - k + 1$ .`
- `dep` is Lemma 1, proved by the elimination of its proof (`elimV`, `skipCoord`). `cnt`, `pairWt_eq` and `sumTo_bound` give Lemma 2 and the exchange of sums. `no_mds_ternary`: for $2 \leq k$, $k + 3 \leq n$ and every G , $\neg \text{MinDistGe } n \ k \ G \ (n - k + 1)$. This is Theorem 3, proved for all n, k symbolically, not by enumeration.
- `IsPAut`, `IsMAut`: permutations (lists of images) and monomial maps sending every codeword to a codeword. `PAutOrder n k G N` and `MAutOrder n k G N` say that a duplicate-free list of length N consists of exactly the automorphisms.
- `Conjecture1090` (and `Conjecture1090Mon`) is the conjunction (`attained 660`) $\wedge$ (`all orders  $\leq 660$`) $\wedge$ (`other orders divide 132`). `conjecture_00000001090_false :  $\neg \text{Conjecture1090}$` and `conjecture_00000001090_monomial_false` refute them through `attained_false`. `no_mds_12_6`, `no_dist7_12_6`, `no_mds_12_6.with` (any property P) and `no_mds_12_7` cover the readings above.
- Non-vacuity: `tetra_mds : IsMDS 4 2 tetra` and `golay_12_6_6` (the Golay code is a $[12, 6]$ code with minimum distance exactly 6; checked over all 729 messages by `decide +kernel`).
- “MDS” dropped: `pair_gen`, `pair_aut`, `pairAuts_nodup`, `atMost660_false` (some $[12, 6]$ code has more than 660 distinct permutation automorphisms), and `pair_order_ge` (if `PAutOrder 12 6 pairG N` then $N \geq 768$). The exact value 46080 is checked in `verify.py` only.

`lake build` succeeds with no errors or warnings. There is no `sorry`, no `native_decide` and no added axiom. `#print axioms` shows only `propext` and `Quot.sound` for all results about MDS codes. The two results on the pair code (`atMost660_false`, `pair_order_ge`) also use the standard axiom `Classical.choice`, through core list lemmas.

6 Independent check

`verify.py` (Python 3 standard library) uses different methods:

- It searches all 1280^2 pairs of words of weight ≥ 7 in $\mathbb{F}_3^8$ and finds none spanning a code with all nonzero weights ≥ 7 , so no $[8, 2, 7]$ code exists. The best ternary $[8, 2]$ code has $d = 6$.
- It checks that no systematic $[5, 2, 4]$ or $[6, 3, 4]$ ternary code exists. An MDS code is always systematic on its first k coordinates.
- It evaluates the Griesmer bound ($n \geq 14$ for $[n, 6, 7]_3$).
- On 300 random $[12, 6]$ ternary codes, it runs the construction of Theorem 3 and computes the exact minimum distance (always ≤ 6).

- It computes the weight distributions of the tetracode and the Golay code.
- It enumerates all 46080 pair-preserving permutations, confirms that they are automorphisms of P , confirms that the weight-2 supports are the six pairs, and re-checks the 768 permutations used in Lean.

7 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
cd .. && python3 verify.py
pdflatex report.tex && pdflatex report.tex
```

References

- [1] F. J. MacWilliams and N. J. A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland, 1977, Ch. 11 (MDS codes; $n \leq q + k - 1$ for $k \geq 2$).
- [2] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999 (the ternary Golay code and $2.M_{12}$).